

Problem Set 1

Course Name: Modeling and Control	LP / Year: 4 / 2025	Course Code: R0004E
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1. (a) Draw and explain a general block diagram of a feedback control system, which includes reference signal, process noise as well as measurement noise as the inputs, as part of the whole system.
 (b) Explain why a feedback control system applies a negative feedback rather than a positive feedback.
 (c) Give an example of a practical feedback control system, and identify each of its elements correspond to what included in the block diagram.
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2. Given an RLC circuit as shown in Figure 1, with the components $R_1 = 1\Omega$, $R_2 = 1\Omega$, $C = 1F$ and $L = 1H$. The input is $v(t)$ and the output is $i_2(t)$.

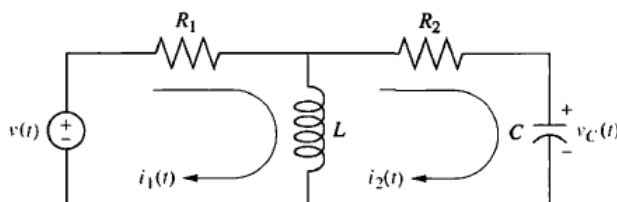


Figure 1: An RLC circuit

- (a) Derive the dynamic equations (differential equations) of the circuit (Hint: use Ohm Law and Kirchoff Laws).
 - (b) Obtain the transfer function $\frac{I_2(s)}{V(s)}$ of the circuit.
 - (c) Find the Laplace inverse of $I_2(s)$ if the voltage input is a unit step $V(s) = 1/s$.
 - (d) Using both (b) and (c), find the final value of $i_2(t)$ if the input $v(t)$ is a unit step. Compare and comment on the two answers.
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3. Consider the mechanical system shown in Figure 2:

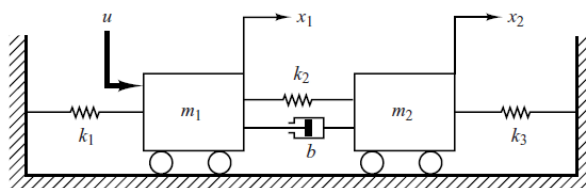


Figure 2: Mass spring damper mechanical system

- (a) Obtain the transfer functions $\frac{X_1(s)}{U(s)}$ and $\frac{X_2(s)}{U(s)}$ of the system.
 - (b) Determine the order of each transfer function you obtained in (a).
 - (c) Determine the (dynamic) variables of the system and the mechanical parameters of the system.
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4. Find the inverse Laplace transform of the following transfer function:

$$\frac{10}{s(s+2)(s+3)^2}$$

5. Given a unit step input, find the steady-state value of the output response of following system:

$$\frac{Y(s)}{U(s)} = \frac{s+1}{s(s^2+4s+8)}$$

6. The dynamical motion of a translational mechanical system is written as a differential equation

$$\ddot{y} + 5\dot{y} + 2y^2 = F$$

Find the linearization of the differential equation at the origin and write the transfer function of the linearized dynamic.

7. A control system consists of four essential elements: plant or process, measurement, evaluation and control. Typical quantities measured in a control system are pressure, level, temperature, flow, position, and speed. Sensors are used to detect and measure those quantities.
- (a) Understanding the definition of system parameters and system variables, explain whether those measure quantities are considered as variables or parameters of the control system.
 - (b) Choose two of the measured quantities. Mention what sensor can be used to measure each of those quantities and explain in a concise way, the working principle of each of the sensors.
 - (c) Explain the differences between a sensor and a transducer.
 - (d) Determine whether each sensor you explain in (b) are also a transducer or not.
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